Adrin Alias

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EDUCATION

University of Texas at Arlington — **Graduation:** December 2025
Bachelor of Science in Mechanical Engineering (Honors) — 3.50 GPA

WORK EXPERIENCE

Tesla — Palo Alto, CA
Mechanical Hardware Test Engineer (Internship) — May 2025 – August 2025

- Engineered an **automated test rig** for the Tesla Semi seat harness using Catia that achieved **100% uptime** over 60k cycles over a large temperature range (-30°C to +85°C), securing primary LV design reliability sign-off.
- Owned the end-to-end validation for a dynamic wire harness, leading test planning, execution, and reporting across design and reliability teams to verify 2x life performance.
- Executed root-cause DOEs to isolate testing noise and implemented hardware fixes (circuit design change) that **reduced noise by 90%**, achieving consistent data across 16 wire configurations.
- Developed a Python data pipeline to visualize **30k+ lines of data** per cycle which was deployed to the team's GitHub to standardize reporting, **cutting post-processing time by 70%**.

Rivian — Irvine, CA
Manufacturing Engineer (Internship) — June 2023 – December 2023

- Identified critical charge port tolerance stack-up issues and negotiated a DFM redesign with Studio to eliminate manual alignment, **saving $55,000** in tooling and removing 13 seconds per vehicle.
- Re-designed an aluminum fixture to align & hold air spring assemblies by analyzing part variation, modeling in Catia, creating part drawings using GD&T, and validating on the line, increasing operator utilization by more than **50%**.

PROJECTS & EXPERIENCE

Brake Dynamometer | Senior Design, Formula SAE — Arlington, TX
Mechatronics Lead *(Hardware, Firmware, & Software)* — August 2024 – May 2025

- Spearheaded architecture of a DAQ for a brake testing rig by overcoming MCU clock limitations using a frequency-to-voltage IC that achieved 20 Hz synchronous sampling at **94% lower cost** than commercial alternatives.
- Developed C++ firmware for a SAME51 microcontroller to process and transmit data from four sensor types (temperature, pressure, force, speed), implementing **CAN bus** communication for a multi-channel IR sensor.
- Designed a Python-based GUI for **real-time data visualization** and logging, enabling the characterization of brake coefficient of friction vs. temperature to validate manufacturer performance and inform design decisions.

IoT Smart Blinds — Arlington, TX
Personal Project — May 2024 – July 2024

- Engineered a **$10** IoT blinds controller (ESP-32) with C++ firmware that uses public weather APIs for sunrise/sunset scheduling. It automatically launches a local web server for manual control upon Wi-Fi failure.
- Designed a fastener-less 3D-printed enclosure (SolidWorks) and validated the system with 30 days of **100% reliability** continuous testing.

Turbomachinery & Energy Systems Laboratory (Research Lab) — Arlington, TX
Mechanical Engineering Assistant — March 2024 – August 2024

- Designed an **8.4 kW** power distribution & control system for 4 heaters, utilizing a digital twin for optimization, custom-built sheet metal enclosures, and PID based control circuits. Cut manufacturing **costs by 30%** while maintaining performance.
- Detected a loss-of-interference failure in a **38K RPM** rotor (Ansys) due to centrifugal expansion. Engineered a geometric redesign to prevent catastrophic failure of the 300lb rig.
- Collaborated with a multidisciplinary team to generate turbine blades by using MATLAB to curve 2D airfoil profiles and SolidWorks to convert them into 3D models.

SKILLS

CAD & Simulation:	SolidWorks, Catia, Ansys Mechanical (FEA), Simulink
Programming:	Python (Data Analysis, Automation), C++ (Embedded/Arduino), MATLAB
Mechatronics:	Circuit Design, Signal Conditioning, PID Control, Motion Control (Actuators, Sensors)
Manufacturing:	GD&T (ASME Y14.5), Rapid Prototyping (3D Printing), Sheet Metal, Laser Cutting, DFM, DFA